

Akash Dubey

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EDUCATION

Rutgers University Honors College, School of Arts and Sciences

New Brunswick, NJ

Bachelor of Science in Computer Science and Mathematics

Anticipated May 2027

- **GPA:** 3.95/4.0 | SAT: 1580
- **Relevant Coursework:** Systems Programming, Computer Architecture, Algorithms, Data Structures, Stochastic Processes, Tensor Networks, Honors Calculus III & IV, Math Theory of Probability.
- **Activities:** Road to Silicon Valley Program (Cohort 6), Quantitative Finance Club, Dean's List, ESL Instructor.

ENGINEERING EXPERIENCE

New York Life Insurance

New York, NY

Software Engineer Intern, TDAV Team

May – Aug 2026

- Built an AI text and voice agent for processing insurance claims for clients and 12,000 Customer Service Representatives on AWS infrastructure with access to NYL services, knowledge, and forms.
- Constructed a company wiki/knowledge base from 13,000 documents compiled from Sharepoint, Public Site, Confluence, Jira, and the Intranet, consolidating business logic into one place for AI and human users.
- Worked on multi-agent systems hosted on NYL's Cloud responsible for improving website development in AEM, Pega code, Tungsten document processing, and mainframe modernization.

Lykke

San Francisco, CA

Founding Software Engineer

Sep 2025 – Present

- Created a distributed course-wiki generation engine that crawls educational sources, synthesizes structured sections, generates interactive artifacts, embeds wiki content for retrieval, and runs tenant-isolated background builds via SQS, MongoDB leases, Redis rate limiting, and AWS ECS.
- Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, Gemini Files API, and SSE streaming, including source inference, citation events, token-budgeting for large course materials, and fallback paths for extracted text vs. full-file reasoning.
- Built core AI workflows for an AI study platform used to ingest Canvas LMS courses, assignments, announcements, notes, and uploaded documents into class-scoped knowledge bases for contextual chat and study generation.

Scarlet Sync

New Brunswick, NJ

Founder

Jan 2025 – Present

- Architected a robust data ingestion pipeline using Python and LLMs to reverse-engineer legacy university data, indexing over 2,500 classes and 500+ degree programs for real-time querying.
- Built a Rutgers academic planning platform with thousands of users for degree, schedule, and class planning plus campus discovery, with a production AI assistant featuring streaming tool-calling and grounded retrieval across Web, ChatGPT, Claude, and iOS.

Samaritan Scout

Cranford, NJ

Full Stack Engineer Intern

May 2023 – Aug 2025

- Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract data from over 100,000 non-profit websites, reducing manual data entry time by over 90%.
- Built a full-stack search engine using React, TypeScript, and PostgreSQL to index 35,000+ volunteer opportunities, implementing semantic search algorithms with hallucination filters to ensure accurate matching.

RESEARCH

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS

New Brunswick, NJ

Research Assistant (Advisor: Prof. Jingjin Yu)

Jan 2026 – Present

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional ASAO robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy (GaP) for dual-arm task-and-motion planning (TAMP), learning from successful and failed planner traces to rank symbolic action skeletons and predict geometric feasibility, working in Isaac Sim to bridge TAMP capabilities with ML.

Kwan Lab, Rutgers Cancer Institute of New Jersey

New Brunswick, NJ

Computational Biology Research Assistant (Advisor: Prof. Kelvin Kwan)

Jan 2026 – Present

- Configured and validated a reproducible DeepSTARR inference environment, resolving legacy TensorFlow/Keras dependencies on HPC to enable sequence-based prediction of enhancer activity.
- Built a 3D chromatin polymer simulator using overdamped Langevin dynamics and multi-scale TAD detection, validating the pipeline with 70+ automated tests.

Rutgers Economics Labs

New Brunswick, NJ

President (formerly Research Director, Team Lead)

Oct 2024 – Present

- Architected RAIL (Rutgers Agentic Intelligence Labs), an AI-automated research pipeline to collect, organize, analyze, and synthesize economic data using data ontologies and coding-agent workflows.
- Leading quantitative research projects for partners including NJDOL, NJDEP, NJBPU, Federal HUD, and the U.S. Census Bureau, applying econometric models in Python/R to analyze large datasets.

SELECTED TECHNICAL PROJECTS

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC, Quantization*

- Deployed models on the Rutgers High Performance Computing (Amarel) cluster, utilizing quantization and structural pruning to maximize inference throughput on constrained compute.
- Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset to study the effects of different model sizes and architectures on math reasoning capabilities.

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Coding Agents*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, markdown graphs, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.
- Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows, including local runner dispatch, dry-run work orders, session logs, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent capable of autonomous computer use by integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA.
- Implemented the Model Context Protocol (MCP) to orchestrate specialized sub-agents (UI Interaction, Shell, App Research) for planning and executing complex multi-app workflows.

AWARDS & SKILLS

Technologies: AWS (SQS, ECS, S3), Redis, MongoDB, PostgreSQL, Distributed Systems, HPC Clusters, PyTorch, Transformers, Pandas, Scikit-learn

Languages: Python, C++, CUDA, Java, SQL, TypeScript, R, Bash, Git

Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.